COUNTEREXAMPLES TO ZHU'S $\mathfrak{S}_N$-LOG-CONCAVITY CONJECTURE

Abstract. We disprove Conjecture 4.3 of Zhu on the $\mathfrak{S}_n$-log-concavity of set-partition orbit harmonics. For every odd $k \geq 5$, set $n = k^2 - 5$ and $d = n - k$. If $k+1 \leq m \leq n$, then the sign representation has multiplicity 4 in $R(\Pi_{n,m})_{d-1} \otimes R(\Pi_{n,m})_{d+1}$, while its multiplicity in $R(\Pi_{n,m})_d^{\otimes 2}$ is 1 for $k = 5$ and 0 for $k \geq 7$. Thus the equivariant injection required by log-concavity does not exist. The first member of the family has $n = 20$.

For $1 \leq m \leq n$, let $\Pi_{n,m}$ be the set of unordered set partitions of $[n]$ whose blocks have size at most m, and write

$$R_{n,m,d} = R(\Pi_{n,m})_d, \qquad X_{n,m,d} = \{P \in \Pi_{n,m} : |P| = n - d\}.$$

Zhu proved the $\mathfrak{S}_n$-module isomorphism

$$R_{n,m,d} \cong \mathbb{C}[X_{n,m,d}] \tag{1}$$

and conjectured that $R(\Pi_{n,m})$ is $\mathfrak{S}_n$-log-concave for $n \geq 9$ [1, Proposition 3.29 and Conjecture 4.3]. In particular, log-concavity would require $\mathfrak{S}_n$-module injections

$$R_{n,m,d-1} \otimes R_{n,m,d+1} \hookrightarrow R_{n,m,d}^{\otimes 2}$$

for every interior degree d.

Theorem 1. *Let $k \geq 5$ be odd, set $n = k^2 - 5$ and $d = n - k$, and suppose that $k + 1 \leq m \leq n$. Then*

$$[\mathrm{sgn} : R_{n,m,d-1} \otimes R_{n,m,d+1}] = 4,$$

whereas

$$[\mathrm{sgn} : R_{n,m,d}^{\otimes 2}] = \begin{cases} 1, & k = 5, \\ 0, & k \geq 7. \end{cases}$$

Consequently, $R(\Pi_{n,m})$ is not $\mathfrak{S}_n$-log-concave, and Conjecture 4.3 of Zhu is false.

The case $k = 5$ gives $n = 20$, $d = 15$, and every $6 \leq m \leq 20$. No minimality in n is claimed.

Lemma 2. *For a finite $\mathfrak{S}_n$-set Y,*

$$[\mathrm{sgn} : \mathbb{C}[Y]] = \#\{\mathcal{O} \in \mathfrak{S}_n \backslash Y : \mathrm{Stab}(y) \subseteq \mathfrak{A}_n \text{ for } y \in \mathcal{O}\}.$$

Proof. For an orbit $\mathcal{O} \cong \mathfrak{S}_n / H$, Frobenius reciprocity identifies

$$\mathrm{Hom}_{\mathfrak{S}_n}(\mathrm{sgn}, \mathbb{C}[\mathcal{O}]) \cong \mathrm{Hom}_H(\mathrm{sgn}|_H, \mathbf{1}).$$

This space has dimension one if $H \subseteq \mathfrak{A}_n$ and zero otherwise. Summing over the orbits proves the formula. $\square$

For an automorphism (σ, π) of a bipartite graph H, write $\operatorname{sgn}_{E(H)}(\sigma, \pi)$ for the sign of its induced permutation of $E(H)$. Let H° be obtained from H by deleting its isolated vertices. All graph isomorphisms below preserve the two vertex classes, and P_j denotes the path on j vertices.

Lemma 3. *The following classifications hold.*

(1) *If Z has four edges, then every automorphism of Z acts evenly on $E(Z)$ if and only if Z° is one of*

$$C_4, \qquad P_5^{(3,2)}, \qquad P_5^{(2,3)}, \qquad P_4 \sqcup K_2,$$

where the superscript records the two bipartition sizes.

(2) *If $Z \subseteq K_{5,5}$ has five edges, then*

$$(*) \qquad \operatorname{sgn}_{E(Z)}(\sigma, \pi) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\pi)$$

for every automorphism (σ, π) of Z if and only if

$$Z^\circ = K_{1,2} \sqcup K_{2,1} \sqcup K_2.$$

Proof. For the first assertion, a connected four-edge bipartite graph is C_4 or a tree on five vertices. The two nonpath trees have two leaves with a common neighbor; transposing those leaves induces one edge transposition. The reflection of P_5 exchanges two pairs of edges, and the side-preserving automorphism group of C_4 is generated by row and column swaps, each of which exchanges two pairs of edges.

If Z is disconnected, its component edge counts are $3+1$, $2+2$, $2+1+1$, or $1+1+1+1$. In the first case the three-edge component is P_4 or $K_{1,3}$, and only $P_4 \sqcup K_2$ survives. Every other case contains a P_3 or $K_{1,3}$ with an odd leaf transposition, or has two K_2 components whose interchange is odd on the edges.

For the second assertion, two isolated vertices on the same side may be transposed without moving an edge, contradicting $(*)$. Hence Z uses at least four vertices on each side. If it uses $(5,5)$ vertices, then $Z = 5K_2$. If it uses $(5,4)$ or $(4,5)$ vertices, then, up to reversing the sides, $Z = K_{1,2} \sqcup 3K_2$. Both graphs fail $(*)$ by interchanging two K_2 components.

It remains to consider used bipartition sizes $(4,4)$. Each side then has degree sequence $(2,1,1,1)$. Let u and v be the degree-two vertices. If uv is an edge, then $Z = P_4 \sqcup 2K_2$, which again fails by interchanging the two K_2 components. If uv is not an edge, then $Z = K_{1,2} \sqcup K_{2,1} \sqcup K_2$. Its automorphism group is generated by the two leaf transpositions. Each generator induces one edge transposition and one transposition in exactly one vertex class, so $(*)$ holds. □

Proof of Theorem 1. By (1),

$$R_{n,m,a} \otimes R_{n,m,b} \cong \mathbb{C}[X_{n,m,a} \times X_{n,m,b}].$$

Consider $(P, Q) \in X_{n,m,a} \times X_{n,m,b}$. If blocks $B \in P$ and $C \in Q$ satisfy $|B \cap C| \geq 2$, then a transposition in $B \cap C$ is an odd element of the stabilizer, so the orbit contributes nothing by Lemma 2.

Suppose instead that every intersection has size at most one. The pair (P, Q) is then encoded by a simple bipartite incidence graph G: its left and right vertices are the blocks of P and Q, and its n edges are the elements of $[n]$. The $\mathfrak{S}_n$-orbits are the side-preserving isomorphism classes of these graphs, and the stabilizer is their side-preserving automorphism group acting on $E(G)$. Such an incidence graph has no isolated vertices.

Let G have r left vertices and c right vertices, and let $Z = K_{r,c} \setminus G$. For an automorphism (σ, π), the action on all rc cells gives

$$\operatorname{sgn}_{E(G)}(\sigma,\pi)\,\operatorname{sgn}_{E(Z)}(\sigma,\pi) = \operatorname{sgn}(\sigma)^c \operatorname{sgn}(\pi)^r. \tag{2}$$

For the source, $(a, b) = (d-1, d+1)$, so

$$(r, c) = (k+1, k-1), \qquad |E(Z)| = (k+1)(k-1) - n = 4.$$

Both r and c are even. Hence the right-hand side of (2) is 1, and an orbit contributes precisely when every automorphism acts evenly on $E(Z)$. Lemma 3 gives exactly four such isomorphism classes. Each listed complement has maximum degree two, so its complement G has no isolated vertices. Moreover, every degree of G is at most $k + 1 \le m$. Therefore

$$[\operatorname{sgn} : R_{n,m,d-1} \otimes R_{n,m,d+1}] = 4.$$

For the target, $(a, b) = (d, d)$, so $r = c = k$ and $|E(Z)| = 5$. If $k = 5$, then (2) shows that an orbit contributes exactly when $(*)$ holds. By Lemma 3, there is one such graph. Its maximum degree is two, so its complement is an admissible incidence graph; hence

$$[\operatorname{sgn} : R_{n,m,d}^{\otimes 2}] = 1.$$

Now let $k \ge 7$. A five-edge graph uses at most five vertices on each side, so Z has two isolated vertices on each side. Transpose two isolated left vertices and fix the right side. This automorphism acts trivially on $E(Z)$, while (2) gives

$$\operatorname{sgn}_{E(G)} = (-1)^k = -1.$$

Thus every transverse orbit has an odd stabilizer element. Together with the nontransverse case, Lemma 2 yields

$$[\operatorname{sgn} : R_{n,m,d}^{\otimes 2}] = 0.$$

Finally, complex representations of $\mathfrak{S}_n$ are semisimple. An injection from the source to the target would force every irreducible multiplicity in the source to be at most the corresponding multiplicity in the target, contrary to the sign multiplicities above. Since the source multiplicity is nonzero, both adjacent graded pieces are nonzero, so d is an interior degree. This proves the theorem. □

References

[1] H. Zhu, *Plethysm and orbit harmonics*, arXiv:2602.12623v1 [math.CO], 2026.